Southern Cross University

Introductory Pharmacology Week 1 Tutorial

Part 1:
Introduction to the Unit

Ken Wojcikowski (PhD; Pharmacology and Toxicology, UQ School of Medicine)

1

Introduction to the Unit developer and the unit

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Unit Developer: Ken Wojcikowski

- Masters in Clinical Pharmacology
- PhD in Pharmacology / Toxicology (UQ School of Medicine
- 25 years teaching clinical and biochemical units
- SCU and National teaching awards

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Section 2: Student Comments for Ken Wojcikowski

Overall, I am satisfied with the teaching of Ken Wojcikowski in this unit (Comments)

Comment
- Thanks Ken for making a great learning experience!
- Like I said previously, Ken is so brilliant and one of the best teachers I have ever had!
- Really impressed with his strive for what he does, and wants his students to do amazingly. Ken is very approachable.
- Thanks again for showing me I have nothing to feat from pharmacology!
- Thank you for all your help!
- Thanks Ken. You are a great teacher and I have learned a lot from you. Thanks for making pharmacology so fun. I was considering changing to pharmacology mid way through this session – such is the extent of your ability to make this subject interesting!
- Ken delivered this unit extremely well. Everything was laid out in bite sized chunks over a weekly basis. His knowledge on the subject matter is invaluable and he conveys his teachings in a learnable manner. This is my second uni degree across two separate universities and i have never come across a course as well organised as this. Thank you Ken.
- Thank You Ken!! Your support has been amazing. You are a credit to SCU! :)
- great!
- Ken has been one of the best tutors in my nursing degree so far, the way the content is given in this unit is great.
- Ken is so passionate about this subject, and it really shows through his teaching. He puts so much time and effort into lectures/classes and it is evident that he truly cares about teaching. Thanks Ken
- The BEST!
- Extremely satisfied with Ken's teaching approach.
- All UAs, tutors and lecturers should be looking to Ken for inspiration on how to make their units' content "pop". Very grateful to have experienced someone with such passion for passing on their vast knowledge.
- Ken was a great teacher, definitely had our best interest at heart to achieve success.
- I love Ken!!!!!
- YES
- An amazing educator, his passion and joy of teaching is evident.
- Ken is the most brilliant, intelligent, articulate lecturer I have ever had I find myself saying " sym path ohh mim ma net tic" with a familiar Ken drawl, so as to maybe absorb some of his genius I hope I have him as a lecturer in the future Fantastic teacher

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PHAR 2001 Student satisfaction: What to expect

In the past 8 years, since Ken Wojcikowski has been delivering this unit, the mean student satisfaction with the unit has been 4.56 out of 5

Recent Improvements:

- **Cut the length of the case study in half**
- **Decreased the Tutorial questions from 5 to 2**
- **The only quiz is on Module 1**

5

PHAR 2001 Unit performance: What to expect

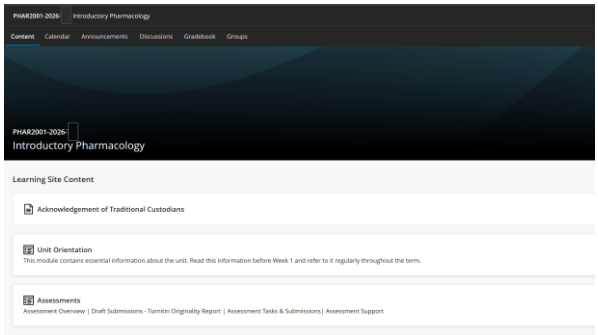
Generally, over 90% of students pass the first time

Most important: Attend tutorials weeks 2, 3, 4 and 5 and engage in Module 1

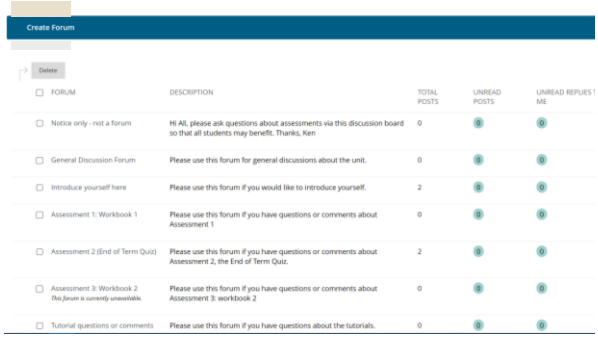
6

Introduction to the PHAR2001 website

7



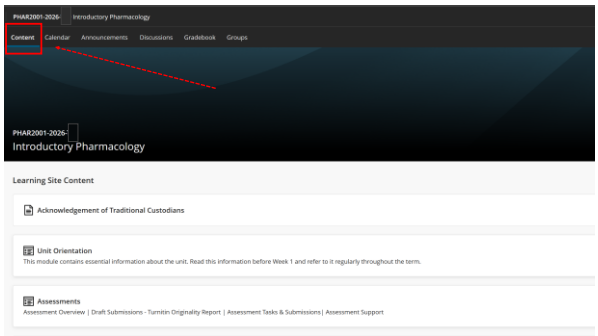
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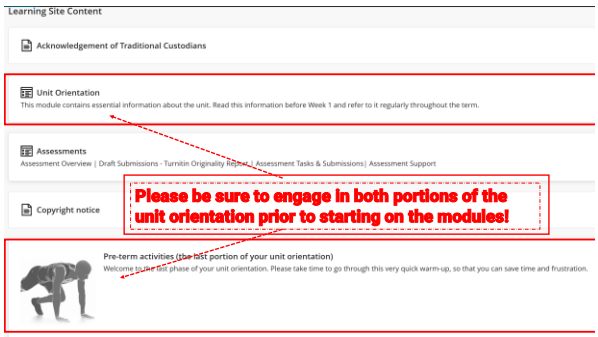
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The unit content:
Your 'Interactive Pharmacology Textbook'

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Let's practice learning from your interactive pharmacology textbook!

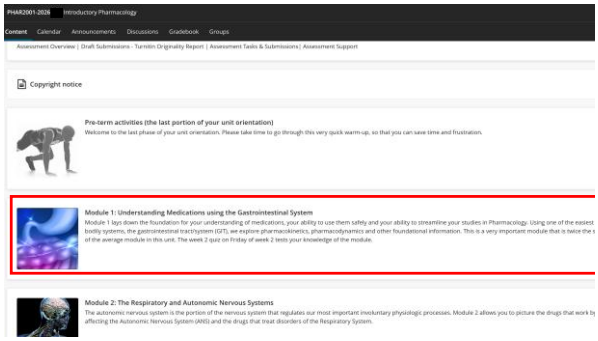
Simply click on the video to play. This video introduces the first three lessons in your interactive pharmacology textbook.



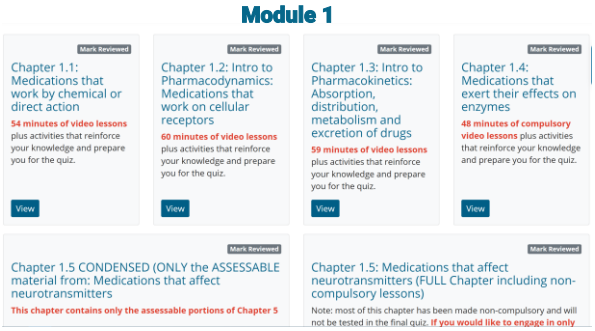
Click below for the notes for the first three lessons and begin your folder of notes that will be invaluable in the final exam.

To maintain academic integrity in the exam given that the final exam is online, all questions in the exam must be worded in a way that students must look up 5 different answers in order to find the correct answer for each question. Students will not have the time to do this for more than a couple questions, so a much better approach will be to make notes on the notes provided and refer to your notes during

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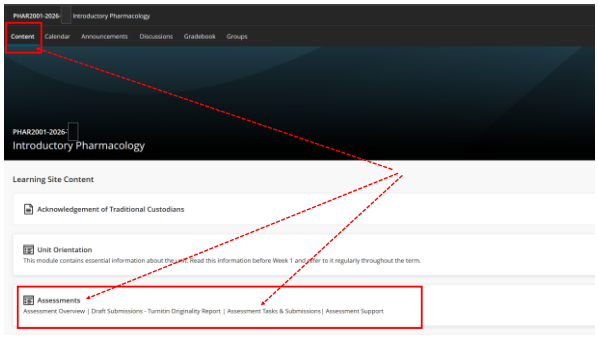
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Introductory Pharmacology Week 1 Tutorial

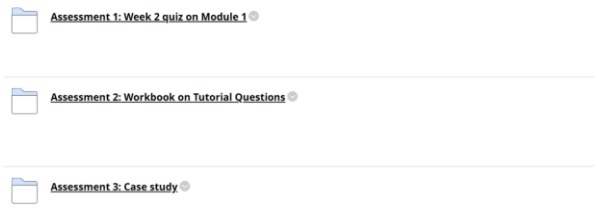
Part 2: Overview of assessments

Ken Wojcikowski (PhD; Pharmacology and Toxicology, UQ School of Medicine)

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Week	Online Learning The online learning prepares you for the quiz and lays down foundational knowledge in pharmacology necessary for your future career. It is your interactive pharmacology textbook.	Online tutorial The tutorials are aimed at helping you do well in the assessments.	Assessment
0 (Orientation week)	**Orientation module**	This module prepares you to work most efficiently within the unit.	
1 26-30 October	**Module 1** 'Understanding medications' using medications of the gastrointestinal system: Pharmacodynamics, Pharmacokinetics and therapeutic foundations.	**Tutorial 1** Developing the research skills to succeed in the case studies (relevant to the case study)	
2 2-6 November	**Module 2** The respiratory and autonomic nervous system medications	**Tutorial 2** Quiz preparation	✓ **Assignment 1: Quiz (20%): Friday 06 November**
3 9-13 November **CENSUS DATE 9 Nov.**	**Module 3** Medications of the endocrine system	**Tutorial 3** Multi-part tutorial question A (relevant to tutorial workbook)	
4 16-20 November	**Module 4** Medications of the cardiovascular system	**Tutorial 4** Multi-part tutorial question B (relevant to tutorial workbook)	
5 23-27 November	**Module 5** Analgesics and medications of the central nervous system	**Tutorial 5** Multi-part tutorial question C (relevant to tutorial workbook)	✓ **Assignment 2: Workbook on tutorial questions is due Wednesday 25 November**
6 30 Nov -4 December	**Module 6** Anti-infective pharmacology	**Tutorial 6** Last chance to address case study questions	✓ **Assignment 3: Case study: due Wednesday 02 Dec.**
7 and beyond	*No further material*		*Grades will be published on 16 Dec*

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Assessment 1: Week 2 quiz

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Week 2 quiz

- **Worth 20% of your grade in the unit**
- **Assesses your knowledge of module 1**
  - **32 Multiple choice questions**
  - **90 minutes**
- **2 – 2-point questions**
  - **The 2-point questions can only be answered if you attend the tutorial in week 2.**

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How to pass the quiz

- Watch the video lessons in Module 1
- Engage in the activities after each lesson
- Go to the Week 2 tutorial, taking good notes
- Re-engage in the activities, adding to your notes

Without a good understanding of the basics of pharmacology, you will not understand future topics, so decide by Census date if you are ready to progress

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Assessment 2:
Workbook:
Multipart tutorial questions
(worth 30%)

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Tips on how to proceed

For ANY credit, you must read the assessment brief and then address each of the sub-questions (in order and numbered) that are contained in the tutorial (they are not in the PowerPoints). Example:

1. ***(Mandatory: PLACE THE QUESTION HERE)***

Please place just the answer to this sub-question here. DO NOT USE POINT FORM. Point form will receive ZERO credit because you do not speak to your client in point form.

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Assessment 3: Case study (worth 50% of your grade)

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7 and beyond	*No further material*		*Grades will be published on 16 Dec*

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The case study

- **You are working at a multidisciplinary clinic**
- **Other healthcare workers are all busy and it is your responsibility to answer questions from clients and colleagues, using the most reliable resources.**
- **The information you provide must be safe, accurate and current**

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Why don't we get the specific resources that answer the case study questions?

- **Assessment 3 does not test your knowledge of the material that we provide you.**
- **It helps you develop the skill of finding reliable safe information about drugs**

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How do we know that we are on the right track with the case study?

- The workshops help answer your questions about the case study.
- Send in your questions by Tuesday, and we will address them in the tutorial
- The discussion board will open up the question to other students.

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Introductory Pharmacology Week 1 Tutorial

Part 3: More about the case study

Ken Wojcikowski (PhD) Pharmacology and Toxicology, UQ School of Medicine

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More details on the Case Studies

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Case study

The case study workbook is worth 50% of your grade in the unit.
It is modelled after those found in your preferred textbook: Bullock, S., & Manias, E. (2022). Fundamentals of Pharmacology. (9th ed.) Pearson 9780655705826. This is a very good Australian/NZ textbook; however, the case study is not directly answered and not fully answered in that text.

Please note that the suggested number of words is not a 'limit'. The 'suggested words' next to each question is the amount of words that the UA took to create a model, full credit answer. Simply use the suggested words as
Please do not exceed 50% greater than the suggested words.

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The case study

- **Has 5 portions, corresponding to the material in Weeks 2, 3, 4, 5 and 6**
- **The modules do not answer the case but help you save time researching because you understand the topics**
- **Make sure you spend a couple hours / week on the case study because**
  - **The material is fresh on your mind**
  - **You can ask questions that we can address in the workshop**

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The case study

- You are working at a multidisciplinary clinic
- Other healthcare workers are all busy and it is your responsibility to answer questions from clients and colleagues, using the most reliable resources.
- The information you provide must be safe, accurate and current

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A word about Academic Integrity

The clips of websites in this tutorial were necessary to guide students on the most appropriate websites and are referenced within the tutorial.

Please remember that it is a breach of academic integrity to copy other's work and use that work as your own.

Please refer to the SCU Academic Integrity Policy for further information.

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2. Is clotrimazole likely to cause any harm to the foetus? Explain your answer. (1 point; about 35 words)

Reference:

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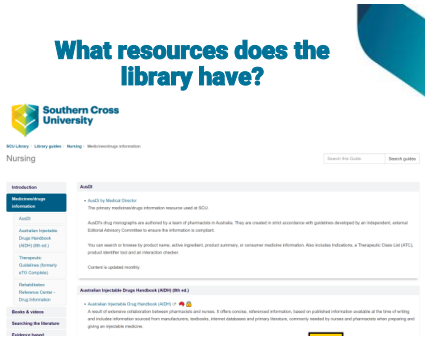
Introductory Pharmacology Week 1 Tutorial

Part 4: Researching the case study

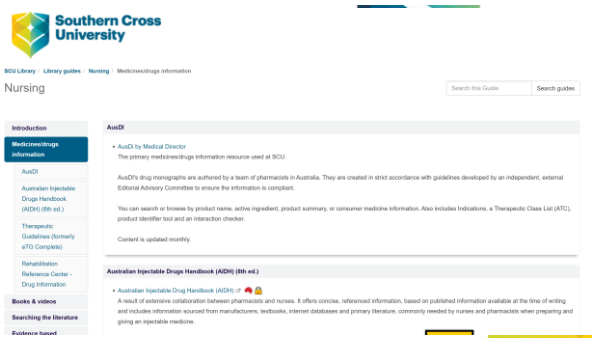
Ken Wojcikowski (PhD) Pharmacology and Toxicology, UQ School of Medicine

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What resources does the library have?



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Specific resources for Researching Case Studies

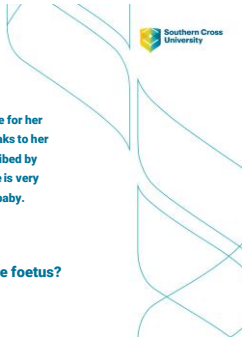
43

Example of a Typical workbook question

HH is a 28-year-old pregnant woman who visits her midwife for her antenatal check-up. She is eight months pregnant and speaks to her midwife about clotrimazole 2 percent vaginal cream prescribed by her local doctor for vulvovaginal candidiasis. ……..she is very concerned about the possible harm of the medicine to her baby.

Question

- Is clotrimazole likely to cause any harm to the foetus? Explain your answer.



44

...just Google it??

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About 2,170,000 results

But which result will contain the information you need?

- ✓*Directed to health professionals*
- ✓*Accurate*
- ✓*evidence based*
- ✓*unbiased*
- ✓*up to date*
- ✓*relevant in Australia?*

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What about Wikipedia?

- May be helpful for a general overview
- **But** you need to check the references
  - Are they from reliable sources?
  - Are they unbiased or are they provided by the medicine sponsor/manufacturer?
  - Publication date?



https://www.wikipedia.org/

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https://www.wikipedia.org/

https://www.wikipedia.org/

https://www.wikipedia.org/

Clotrimazole, sold under the brand name **Lotrimin**, among others, is an antifungal medication.[1] It is used to treat vaginal yeast infections, oral thrush, diaper rash, tinea versicolor, and types of ringworm including athlete's foot and jock itch.[1] It can be taken by mouth or applied as a cream to the skin or in the vagina.[1]

Common side effects when taken by mouth include nausea and itchiness.[1] When applied to the skin, common side effects include redness and a burning sensation.[1] In pregnancy, use on the skin or in the vagina is believed to be safe.[1] There is no evidence of harm when used by mouth during pregnancy but this has been less well studied.[1] When used by mouth, greater care should be taken in those with liver problems.[1] It is in the azole class of medications and works by disrupting the fungal cell membrane.[1]

Clotrimazole was discovered in 1969.[2] It is on the World Health Organization's List of Essential Medicines.[3] It is available as a generic medication.[1]

Medical uses [edit]

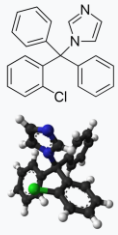
It is commonly available without a prescription in various dosage forms, such as a topical cream, ointment, or vaginal suppository. It is also available as an oral troche or throat lozenge as a prescription only. Topically, clotrimazole is used for vulvovaginal candidiasis (yeast infection) or yeast infections of the skin. For vulvovaginal candidiasis, clotrimazole tablets and creams are inserted into the vagina. Topical clotrimazole is usually not effective in treatment of fungal infections of the scalp or nails.[citation needed] When using over-the-counter drug clotrimazole products, use should be discontinued if condition does not improve after treatment for 2 weeks for jock itch or after 4 weeks for athlete's foot or ringworm.[4]

Throat lozenge preparations are used for oropharyngeal candidiasis (oral thrush) or prevention of oral thrush in people with neutropenia.[4]

Clotrimazole is usually used five times daily for 14 days for oral thrush, twice daily for 2 to 8 weeks for skin infections, and once daily for 3 or 7 days for vaginal infections.[5]

Clotrimazole may be compounded with a glucocorticoid, such as betamethasone, in a topical cream for the treatment of tinea corporis (ringworm), tinea cruris (jock itch) and tinea pedis (athlete's foot). Although FDA

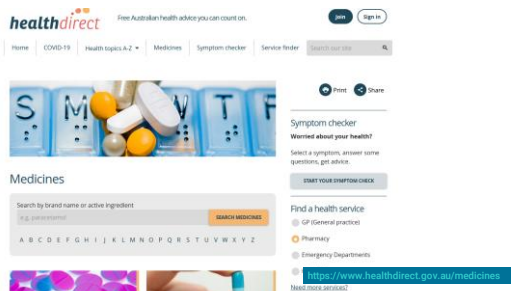
Clotrimazole



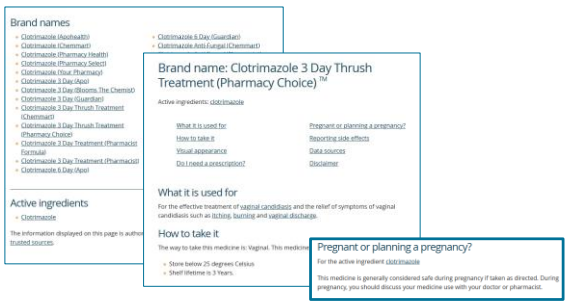
Clinical data	
Trade names	Desenex, CalmYourself others
AHFS/Drugs.com	Monograph
MedlinePlus	a682753
Pregnancy category	AU: A

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Australian government consumer sites ?



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https://www.healthdirect.gov.au/medicines/brand/amt,735751000168105/clotrimazole-apohealth

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Websites for health professionals

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Websites for Health Professionals

NPS MEDICINEWISE

Medicine Finder: provides links to information about many types of medicines

https://www.nps.org.au/medicine-finder

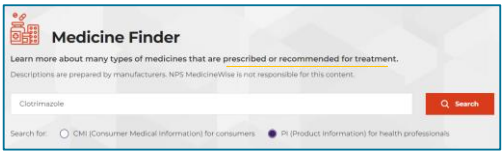
Hosted by the Australian Commission on Safety and Quality in Health Care

Drawbacks

- Descriptions provided by the manufacturer
- Many medicines not yet included in the database

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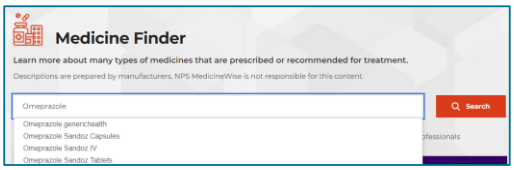
Medicine Finder



Clotrimazole – no results
Canesten – no results

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Proton pump inhibitor – frequently prescribed class of medicines



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5

Drugs.com



This website is from the United States and draws its information from a range of independent sources such as the FDA, The American Society of Health-System Pharmacists and Harvard Health.

The Pro Edition - for healthcare professionals.

- **AHFS DI Monographs – drug information**
- **FDA Professional Drug Information**
- **Drug Interaction Checker**
- **Professional Patient Advice**

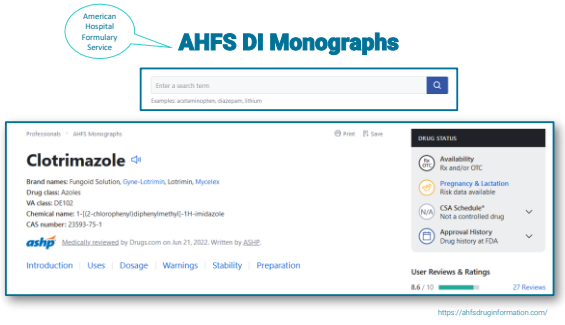
And more

https://www.drugs.com/professionals.html

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AHFS DI Monographs

American Hospital Formulary Service



https://ahfsdruginformation.com/

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AHFS Monograph on Clotrimazole

Introduction

Antifungal; azole (imidazole derivative).

Uses for Clotrimazole

keep on scrolling to find the condition in the case study...

Vulvovaginal Candidiasis

Treatment of uncomplicated vulvovaginal candidiasis (mild to moderate, sporadic or infrequent, most likely caused by *Candida albicans*, occurring in immunocompetent women).

https://www.drugs.com/monograph/clotrimazole.html

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AHFS Monograph Clotrimazole Dosage and Administration

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Administration

Administer topically as oral lozenge; to skin as cream, lotion, or solution; or intravaginally as cream or tablet.

Topical cream, lotion, and solution not intended for ophthalmic use.

Topical preparations containing clotrimazole in fixed combination with betamethasone dipropionate not intended for ophthalmic, oral, or intravaginal use.

Oral Topical Administration

Dissolve oral lozenges slowly in mouth over approximately 15–30 minutes.

Topical Administration

Apply cream, lotion, or solution sparingly in the morning and evening; rub gently into cleansed, affected area and surrounding skin.

Intravaginal Topical Administration

Administer preferably at bedtime.

https://www.drugs.com/monograph/clotrimazole.html

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AHFS Monograph Cautions for Clotrimazole

Specific Populations

Pregnancy

Category B (topical and intravaginal preparations).

Category C (oral lozenges; topical preparations containing betamethasone dipropionate).

CDC and others state that a 7-day regimen of an intravaginal azole antifungal can be used, if necessary, for treatment of vulvovaginal candidiasis in pregnant women.

https://www.drugs.com/monograph/clotrimazole.html

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AHFS DI Monographs

Professionals › AHFS Monographs

Print Save

Clotrimazole

Brand names: Fungoid Solution, Gyne-Lotrimin, Lotrimin, Mycelex
Drug class: Azoles
VA class: DE102
Chemical name: 1-[(2-chlorophenyl)diphenylmethyl]-1H-imidazole
CAS number: 23593-75-1

ashp Medically reviewed by Drugs.com on Jun 21, 2022. Written by ASHP.

Introduction | Uses | Dosage | Warnings | Stability | Preparation

DRUG STATUS

Availability
Rx and/or OTC

Pregnancy & Lactation
Risk data available

CSA Schedules*
Not a controlled drug

Approval History
Drug history at FDA

User Reviews & Ratings
8.6 / 10 27 Reviews

https://www.drugs.com/monograph/clotrimazole.html

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Clotrimazole topical Pregnancy and Breastfeeding Warnings

Medically reviewed by Drugs.com. Last updated on Feb 28, 2022.

Clotrimazole topical is also known as: Abtrim, Abtrim Pessaries, Canesten, Canesten 1, Canesten 3, Canesten 6 Day Pessary, Clotrimazole-3, Clotrimazole-7, Desenex Antifungal Cream, Femcare, FungiCURE Pump Spray, Gyne-Lotrimin, Gyne-Lotrimin 3 Day, Gyne-Lotrimin Inserts, Lotrimin AF Athlete's Foot Cream, Lotrimin AF For Her Athlete's Foot Cream, Lotrimin AF Jock Itch Cream, Lotrimin AF Ringworm Cream, Mycelex, Mycelex OTC, Trivagizole 3

Pregnancy Warnings | Breastfeeding Warnings

Clotrimazole topical Pregnancy Warnings

Clotrimazole topical has been assigned to pregnancy category B by the FDA. Systemic absorption is minimal following topical or vaginal administration. Animals given high doses intravaginally have not revealed evidence of teratogenicity. High oral doses resulted in embryotoxicity and maternal toxicity. In clinical trials involving women in their second and third trimester, clotrimazole treatment of vaginal candidiasis was not associated with fetal harm. Clotrimazole use has not been well studied during the first trimester. Clotrimazole should only be used during the first trimester when need has been clearly established.

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Therapeutic Goods Administration TGA

The Australian categorisation system for prescribing medicines in pregnancy was developed for the TGA by a group of medical and scientific experts based on available evidence of risks associated with taking a particular medicines while pregnant.

A searchable database is available

Search: clotrimazole Show 10 entries

Drug name	Category	Safety statement	Classification level 1	Classification level 2	Classification level 3
clotrimazole	A	-	Genitourinary System	Topical vaginal medication	-
clotrimazole	A	-	Drugs used in Dermatology	Topical antifungals, antiseptics (see also antifungal agents)	-

https://www.tga.gov.au/australian-categorisation-system-prescribing-medicines-pregnancy

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Classifying drugs in pregnancy:

The Australian system

The Australian categories for prescribing medicines in pregnancy [1]

Category A

Drugs which have been taken by a large number of pregnant women and women of childbearing age without any proven increase in the frequency of malformations or other direct or indirect harmful effects on the fetus having been observed.

Category B1

Drugs which have been taken by only a limited number of pregnant women and women of childbearing age, without an increase in the frequency of malformation or other direct or indirect harmful effects on the human fetus having been observed.

Studies in animals have not shown evidence of an increased occurrence of fetal damage.

Category B2

Drugs which have been taken by only a limited number of pregnant women and women of childbearing age, without an increase in the frequency of malformation or other direct or indirect harmful effects on the human fetus having been observed.

Studies in animals are inadequate or may be lacking, but available data show no evidence of an increased occurrence of fetal damage.

Category B3

Drugs which have been taken by only a limited number of pregnant women and women of childbearing age, without an increase in the frequency of malformation or other direct or indirect harmful effects on the human fetus having been observed.

Studies in animals have shown evidence of an increased occurrence of fetal damage, the significance of which is considered uncertain in humans.

Kennedy D. Classifying drugs in pregnancy. Aust Prescr 2014;37:38-40 https://doi.org/10.18773/austprescr.2014.018

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Classifying drugs in pregnancy:

The Australian system

Category C

Drugs which, owing to their pharmacological effects, have caused or may be suspected of causing, harmful effects on the human fetus or neonate without causing malformations. These effects may be reversible. Accompanying texts should be consulted for further details.

Category D

Drugs which have caused, are suspected to have caused or may be expected to cause, an increased incidence of human fetal malformations or irreversible damage. These drugs may also have adverse pharmacological effects. Accompanying texts should be consulted for further details.

Category X

Drugs which have such a high risk of causing permanent damage to the fetus that they should not be used in pregnancy or when there is a possibility of pregnancy.

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The TGA prescribing medicines in pregnancy database

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Search: clotrimazole Show 10 entries

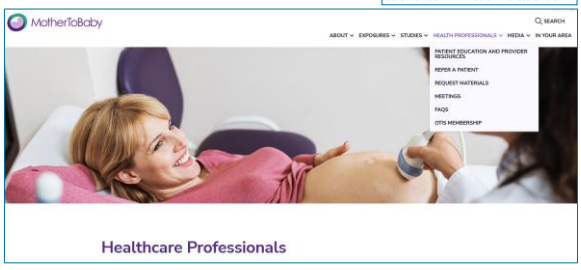
Drug name	Category	Safety statement	Classification level 1	Classification level 2	Classification level 3
clotrimazole	A	-	Genitourinary System	Topical vaginal medication	-
clotrimazole	A	-	Drugs used in Dermatology	Topical antifungals, antiseptics (see also antifungal agents)	-

https://www.tga.gov.au/australian-categorisation-system-prescribing-medicines-pregnancy

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MotherToBaby

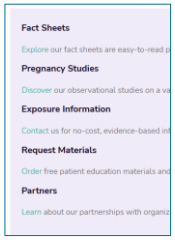
Scroll on down in Health Professionals



https://mothertobaby.org/healthcare-professionals/

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Access to MotherToBaby Factsheets



- **Click Explore to take you to the webpage**
- **Use the link to get directly into the factsheets**

https://mothertobaby.org/fact-sheets/

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MotherToBaby | FACT SHEET

Miconazole | Clotrimazole

This sheet is about exposure to miconazole or clotrimazole in pregnancy and while breastfeeding. This information should not take the place of medical care and advice from your healthcare providers.

What is miconazole?

Miconazole is a medication used to treat fungal infections. Miconazole is a cream used to treat skin and vaginal infections. Common trade names for miconazole include Monistat®, Micatin® and Mitrazole®.

What is clotrimazole?

Clotrimazole is also a medication used to treat fungal infections. Like miconazole, clotrimazole is available as a cream to treat skin and vaginal infections. Brand names of clotrimazole Femcare®, Lotrimin®, and Mycelex®.

I just found out I am pregnant. Should I stop using miconazole or clotrimazole?

Sometimes when people find out they are pregnant, they think about changing how they take their medication, or stopping their medication altogether. However, it is important to talk with your healthcare providers before making any changes to how you take this medication. Your healthcare providers can talk with you about the benefits of treating your condition and the risks of untreated illness during pregnancy.

What should I do if I think I have a vaginal yeast infection while pregnant?

People are more likely to get yeast infections during pregnancy than at other times. If you think you have a vaginal yeast infection, it is important to see your healthcare provider to be sure the infection is yeast before trying to treat it on your own. If you have another type of infection, you may need different treatment.

I use miconazole or clotrimazole. Can it make it harder for me to become pregnant?

Using miconazole or clotrimazole is not expected to make it harder to become pregnant.

Does using miconazole or clotrimazole increase the chance for miscarriage?

https://mothertobaby.org/fact-sheets/miconazoleclotrimazole-pregnancy/

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Does using miconazole or clotrimazole increase the chance for miscarriage?

Miscarriage can occur in any pregnancy. One study found a small increased chance for miscarriage with miconazole and clotrimazole, but there were several problems with this study that could have affected the results. Other studies have not found that miconazole or clotrimazole increases the chance for miscarriage.

Does taking miconazole or clotrimazole increase the chance of birth defects?

Every pregnancy starts out with a 3-5% chance of having a birth defect. This is called the background risk. Topical (used on the skin) or vaginal medications enter the body in lower amounts than oral (pill) medications. This means less medication reaches the developing baby. Since topical and vaginal miconazole and clotrimazole are not well absorbed, they are unlikely to be a concern for the pregnancy. Most studies have shown that miconazole or clotrimazole at low doses (<400 mg/day) does not increase the chance of birth defects.

Does using miconazole or clotrimazole in pregnancy affect future behavior or learning for the child?

Studies have not been done to see if miconazole or clotrimazole can cause behavior or learning issues for the child.

Breastfeeding while using miconazole or clotrimazole:

There are currently no studies looking at miconazole or clotrimazole use during breastfeeding. However, because only small amounts of the medication could pass into breastmilk when miconazole or clotrimazole is used topically, breastfeeding is not expected to be a concern. Miconazole and clotrimazole creams have been used directly on infants under the care of a healthcare provider to treat fungal infections. Talk to your healthcare provider about your breastfeeding questions.

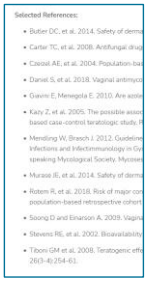
https://mothertobaby.org/fact-sheets/miconazoleclotrimazole-pregnancy/

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Access to MotherToBaby Factsheets

- Is this the best resource for the assignment or to inform a health professional?

- **Lacks the drug classification details**
- **Hard to judge how up to date it is as the link provides only selected references.**
- **More recent research available**


Selected References:

- Butler DC, et al. 2014. Safety of derma
- Carter TC, et al. 2008. Antifungal drug
- Czeizel AE, et al. 2004. Population-ba
- Daniel S, et al. 2018. Vaginal antimyco
- Gianni E, Menegola E. 2010. Are azole
- Kaity Z, et al. 2005. The possible asso based case-control teratologic study. P
- Mendling W, Brasch J. 2012. Guideline Infections and Infectiommunology in Gy speaking Mycological Society. Mycoses
- Murase JE, et al. 2014. Safety of derma
- Rotem R, et al. 2018. Risk of major con population-based retrospective cohort
- Soong D and Einarson A. 2009. Vagina
- Stevens RE, et al. 2002. Bioavailability
- Tiboni GM et al. 2008. Teratogenic effe 26(3-4):254-61.

https://mothertobaby.org/fact-sheet-reference/miconazoleclotrimazole/

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Introductory Pharmacology Week 1 Tutorial

Part 5: Peer reviewed research

Ken Wojcikowski (PhD) Pharmacology and Toxicology, UQ School of Medicine

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Peer reviewed research

If all else fails and you have not found what you are looking for, you can check out the peer reviewed search engines PubMed and Google Scholar.

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Sourcing your answer from peer reviewed papers

Latest and greatest sources

- **PubMed**
- **Google Scholar or**
- **SCU library***

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PubMed

PubMed is a collection of millions of biomedical journal citations and abstracts as well as some books/chapters.

It provides links to full text of journal papers that are open access

PubMed is free and global

Content

- **MEDLINE – the National Library of Medicine database**
- **PubMed Central (PMC)**
- **Bookshelf – archive of full text books reports**

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PubMed



https://pubmed.ncbi.nlm.nih.gov/

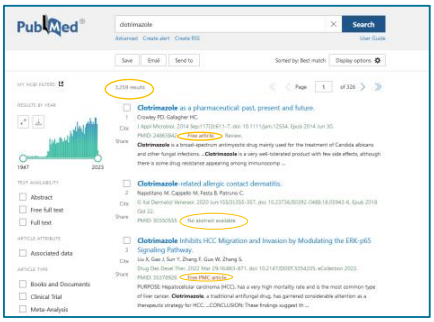
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Simple search

Typing a medication name brings up several search options

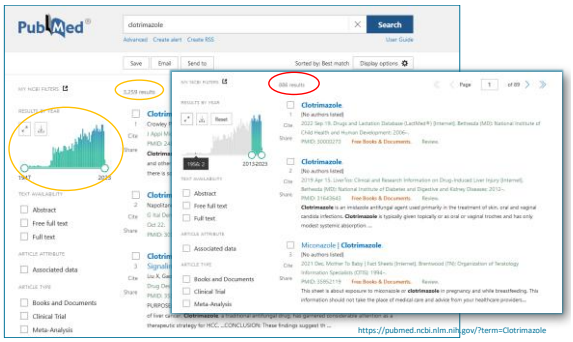


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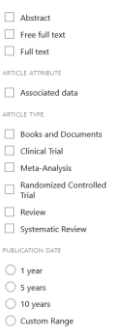
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https://pubmed.ncbi.nlm.nih.gov/?term=Clotrimazole

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More filters

- **Check out the left hand column**
- **Adjust your search fast, by ticking boxes**

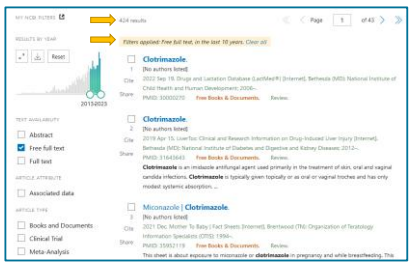
Options

- TEXT AVAILABILITY: limit the search to citations with full text attached, or just free full text
- ARTICLE ATTRIBUTE: rarely relevant
- ARTICLE TYPE: depends on the research question you are answering
- PUBLICATION DATE: focuses your search to a timespan

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Now with two filters …



https://pubmed.ncbi.nlm.nih.gov/?term=Clotrimazole&filter=simsearch2.ffrft&filter=datesearch.y_10

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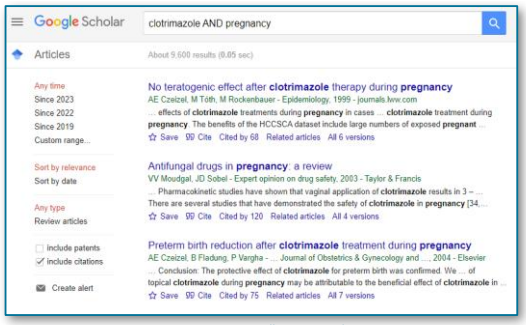
Searching with Google Scholar



https://scholar.google.com.au/ or simply search 'Google Scholar' and click

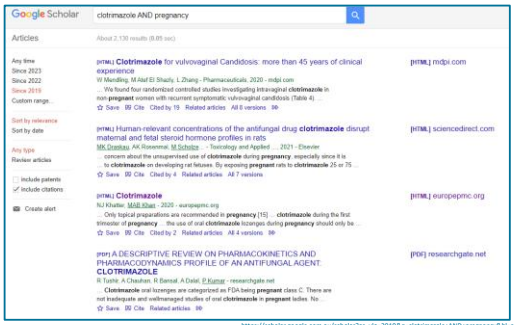
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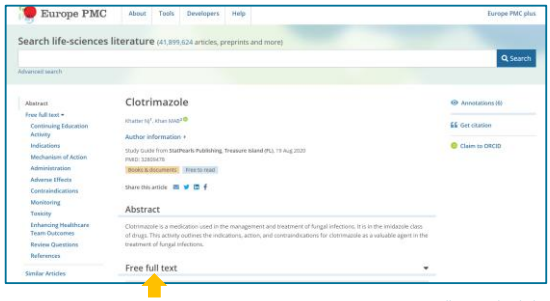
https://scholar.google.com.au/scholar?hl=en&as_sdt=0%2C5&q=clotrimazole+AND+pregnancy&btnG=

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https://scholar.google.com.au/scholar?as_ylo=2019&q=clotrimazole+AND+pregnancy&hl=en&as_sdt=0,5

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https://europepmc.org/article/nbk/nbk560643

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Pregnancy

Clotrimazole demonstrates poor absorption after dermal or intravaginal administration. Only topical preparations are recommended in pregnancy.[15] There are inadequate well-controlled human studies with the use of topical or intravaginal clotrimazole during the first trimester of pregnancy clotrimazole should only be used if indicated.[16] However, clinical trials showed intravaginal clotrimazole to be safe in clinical trials during the second and the third trimester of pregnancy and are recommended for use.[17][12]

Clotrimazole is not known to cross the placenta.[11] FDA classifies clotrimazole as a class C drug in pregnancy risk classification. There are no adequate, well-controlled studies for oral clotrimazole in pregnant women. There have been no teratogenic effects demonstrated after clotrimazole therapy in pregnancy.[17] In animal studies with dosing up to 200 times the human dose, doses of 100 times the adult human dose have been embryotoxic in rats and mice. There are well-controlled studies in humans; hence the use of oral clotrimazole lozenges during pregnancy should only be an option if the potential benefit justifies the potential risk to the fetus.

Breast-feeding

The use of clotrimazole during breastfeeding has not been studied. No data is available about the excretion of clotrimazole in breast milk. Topical clotrimazole poses little risk to nursing infants as topical clotrimazole is not expected to result in significant maternal absorption.[18] Instruct mothers not to apply clotrimazole topically to the breast during times of breastfeeding. The oral troches should be used only when needed, as this may be systemically absorbed. Report to the FDA if a breastfeeding infant experiences any adverse effects.

https://europepmc.org/article/nbk/nbk560643

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Researching questions about Pharmacokinetics and Pharmacodynamics

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Pharmacokinetics
'Pharmaco' (drug or poison)
'kinetics' (movement of)

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Imagine what our body does to the drug...

- **Absorption**
- **Metabolism**
- **Distribution**
- **Excretion**

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Pharmacodynamics

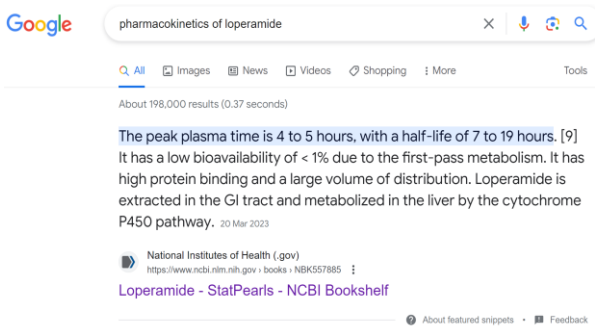
Imagine what the drug does to our body…

- **Does it work on cellular receptors?**
- **Does it work on enzymes?**

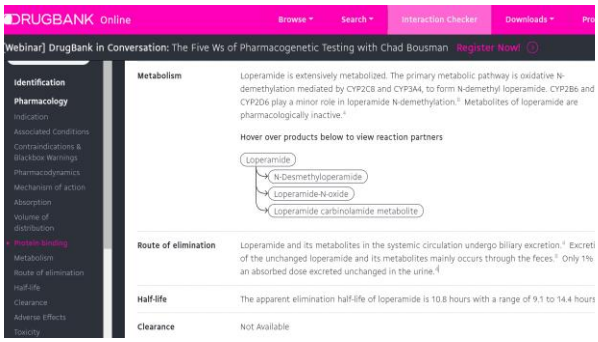
Pharmacodynamics includes the mechanism of action

But we are only concerned about an overview of the mechanism of action – sometimes called 'mode of action'

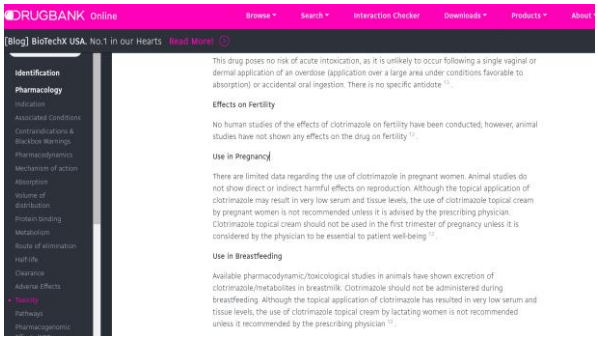
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Thanks for attending!

Remember, our goals in the clinical cases is to develop the skill of investigating complex medical questions about medications and accurately convey that information to clients and colleagues.

Accuracy and safety is of utmost importance.

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References

All clips of websites in this tutorial are referenced within the tutorial.

Please remember that it is a breach of academic integrity to copy other's work and use that work as your own.

Please refer to the SCU Academic Integrity Policy for further information.

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